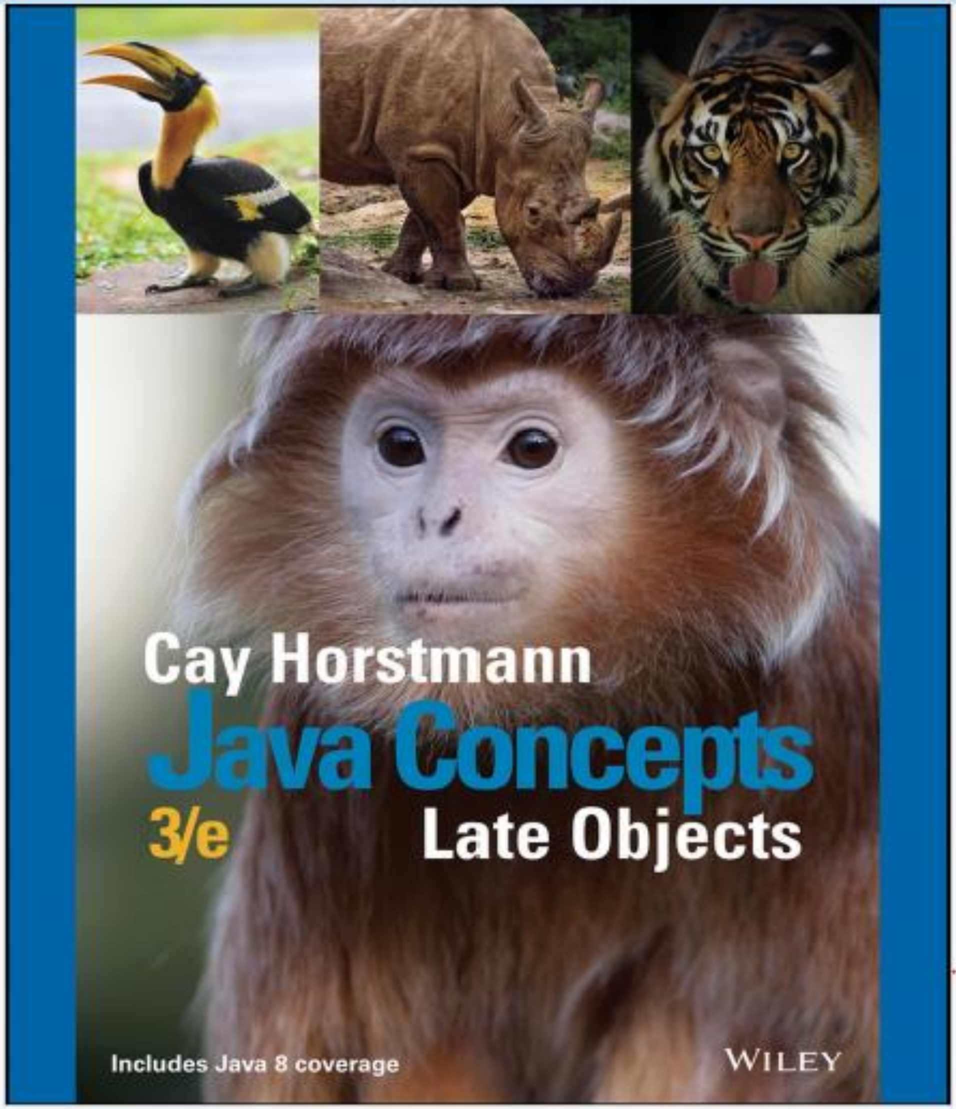


CSI 141 Programming Principles

Lecture 1 Introduction

Objectives

- To learn about computers and programming
- To compile and run your first Java program
- To recognize compile-time and run-time errors
- To describe an algorithm with pseudocode

Outline

- Computer Programs
- The Anatomy of a Computer
- The Java Programming Language
- Programming Environment

Computer Programs

Definitions

- The **computer** itself is a machine that stores data (numbers, words, pictures), interacts with devices (the monitor, the sound system, the printer), and executes **programs**.
- A **program** is a **set of instructions**, that are **written in a language** that the computer understands, to perform a specific task.
- The physical computer and peripheral devices are collectively called the **hardware**.
- The act of designing and implementing computer programs is called **programming**.

Computer Hardware

Computer language

Machine Language

```
01001000 01100101
01101100 01101100
01101111 00100000
01010111 01101111
01110010 01101100
01100100
```

- ✓ **Understood by the computer**
- ✓ **People don't like it at all**

How about natural language?

- ✓ **English?**
- ✓ **Setswana?**

Assembly Language

```
MOV  AX,DATA
MOV  DS,AX
LEA  DX,MSG1
MOV  AH,9
INT  21H
MOV  AH,1
```

- ✓ **Easily understood by the computer**
- ✓ **Has to be converted to machine language**
- ✓ **People can live with it**

High-level Language

```
for(int i = 0; i < items.length; i++){
   System.out.print(items[i]);
   if (i < items.length -1 )
      System.out.print(", ");
}
```

- ✓ **A compromise**
- ✓ **Must be converted to machine language**
- ✓ **At this point, our solution**

Computer language vs Natural language

Parameter	Natural language processing	Programming Language
Purpose	Connected in processing the **human** natural language, one of the sub-categories of AI	Way of writing instructions to the **computer**
Syntax	Generates **human language** syntax	Strict syntax **for every language**
Aim	enable computers to **interact** with human language	**Solve** the task and computational problems and do all manipulation
Works on	Works with **unstructured** and speech data	Works with **structured** data, variables, and program logic
Used by	data scientists, computational linguists and NLP experts	Programmers, software developers
Communication	Focuses on processing and understanding human language text data	Used for specifying algorithms and manipulation
Application	Chatbots, language translation, speech recognition, etc	develop software, applications and algorithms
Examples	machine translation, sentiment analysis	C,C++,java,python etc.
Error management	uses probabilistic models	through try-catch blocks
Tools	NLTK, TensorFlow	IDEs (Integrated Development Environments), compilers

Aug 2025	Aug 2024	Change	Programming Language	Ratings	Change
1	1		Python	26.14%	+8.10%
2	2		C++	9.18%	-0.86%
3	3		C	9.03%	-0.15%
4	4		Java	8.59%	-0.58%
5	5		C#	5.52%	-0.87%
6	6		JavaScript	3.15%	-0.76%
7	8	↑	Visual Basic	2.33%	+0.15%
8	9	↑	Go	2.11%	+0.08%
9	25	↑↑	Perl	2.08%	+1.17%
10	12	↑	Delphi/Object Pascal	1.82%	+0.19%

https://www.tiobe.com/tiobe-index/

Quiz 1

1. What is required to play music on a computer?
2. Why is a CD player less flexible than a computer?
3. What does a computer user need to know about programming in order to play a video game?

computer performs the following tasks:

- **Input:** Sending the data and command to the computer is known as input.
- **Processing:** Work done by the computer with the help of processing hardware and software to produce results is known as processing.
- **Output:** The result displayed by the computer is known as output.
- **Storage:** A place to save result inside or outside the computer is known as storage.

The Anatomy of a Computer

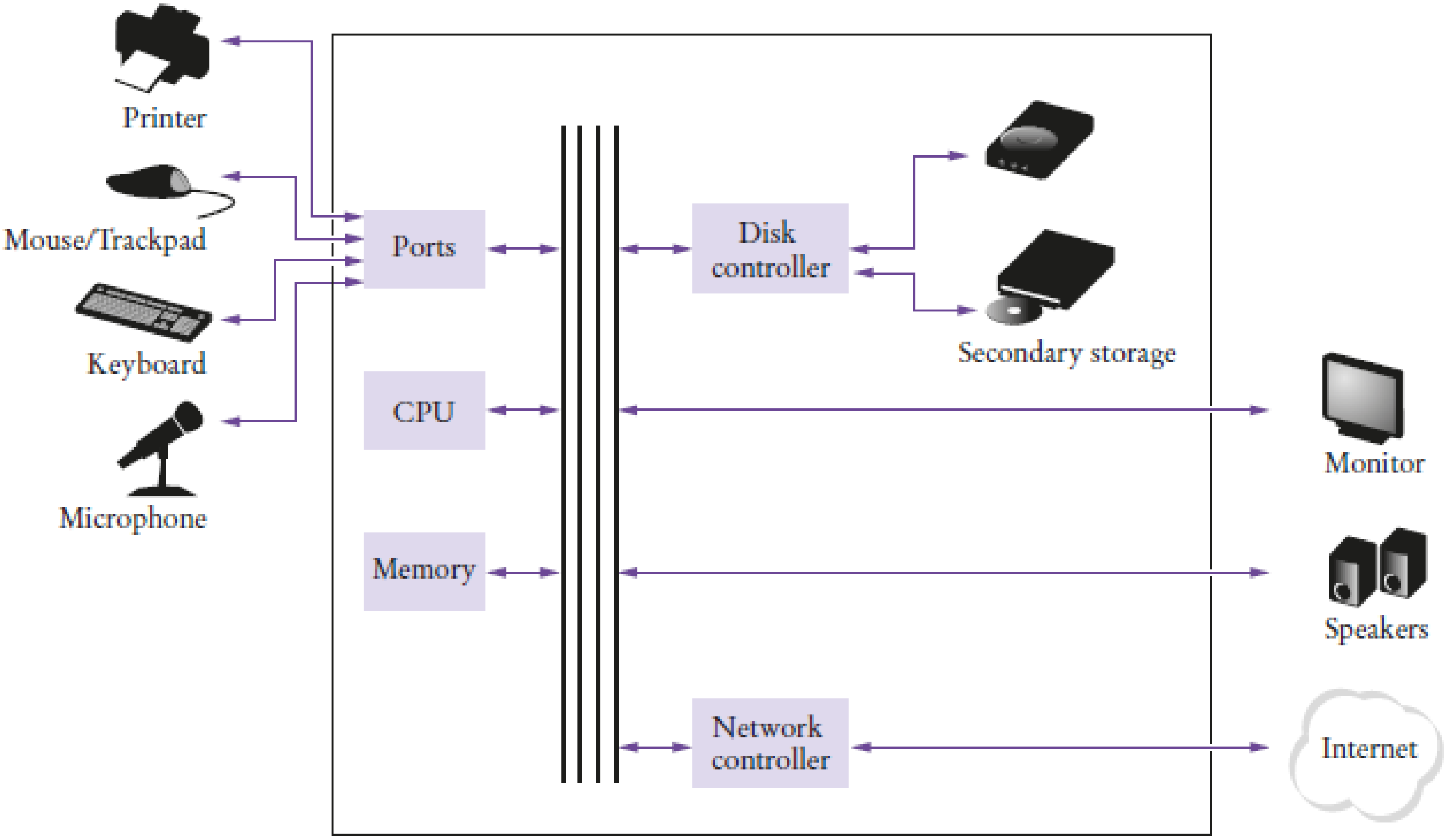


Figure 1: Schematic Design of a Personal Computer

Computer system

The computer system essentially comprises three important parts:

1. input device,
2. central processing unit (CPU)
3. output device.

The Anatomy of a Computer

- core functional units (input, output, memory, arithmetic logic unit (ALU), and control unit) and
- the hardware components (Motherboard, Central Processing Unit (CPU), Random Access Memory (RAM), Hard Drive/Solid State Drive (Storage), Input Devices, Output Devices, Power Supply Unit)

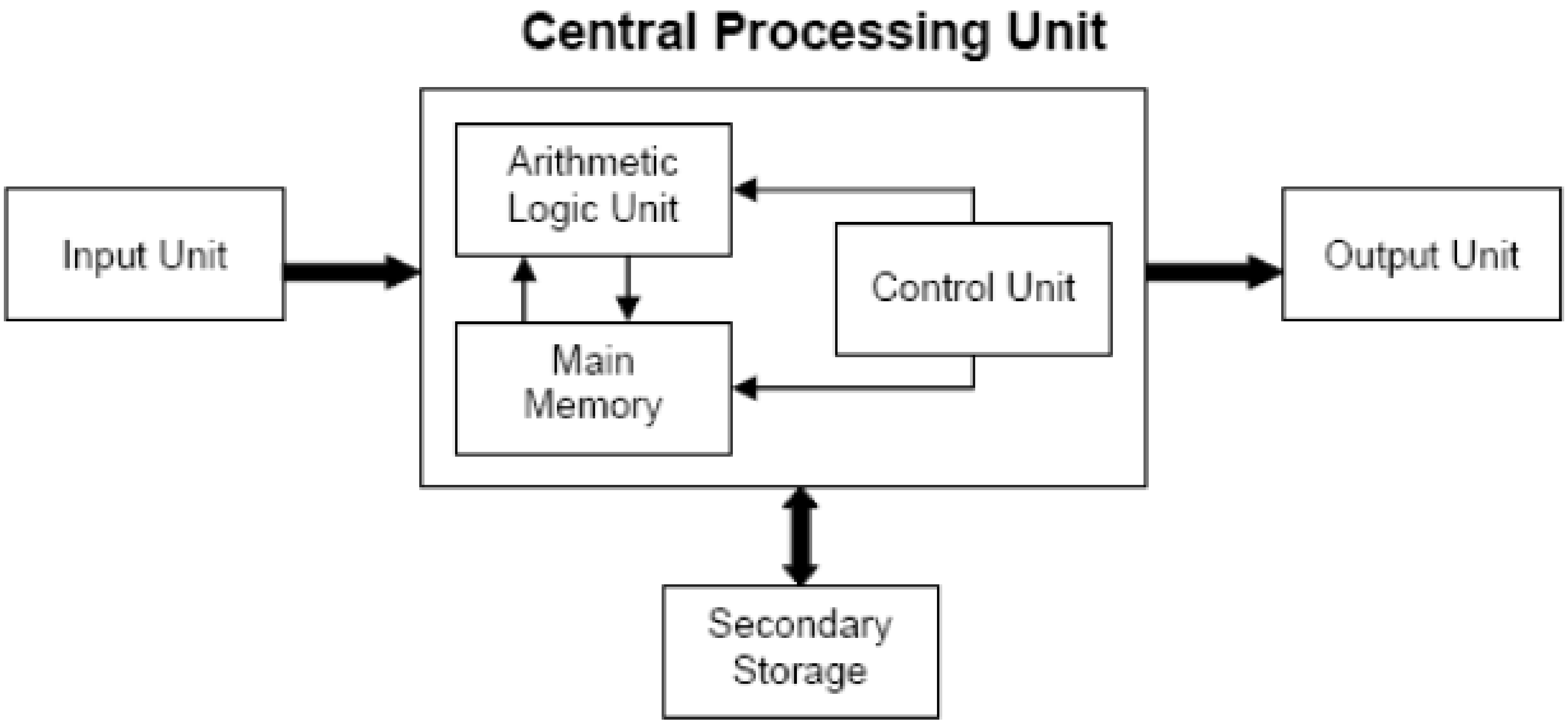
❑ Core functional units

- ❑ **Input Unit**: These allow users to send data and instructions to the computer. Examples : keyboards, mice, and scanners.
- ❑ **Output Unit**: These display or present processed information to the user. Examples : monitors, printers, and speakers.
- ❑ **Memory Unit**: This is where data and instructions are stored for the CPU to access.
- ✓ RAM (Random Access Memory), which is temporary and fast,
- ✓ **storage devices** like hard drives or SSDs, which provide long-term storage.
- ❑ **Central Processing Unit** (CPU): Often called the "brain" of the computer, the CPU is responsible for **executing** instructions and performing **calculations**. It consists of:
- ✓ **ALU (Arithmetic Logic Unit):** Performs arithmetic and logical operations.
- ✓ **Control Unit:** Fetches instructions, decodes them, and directs other components to carry out the instructions.

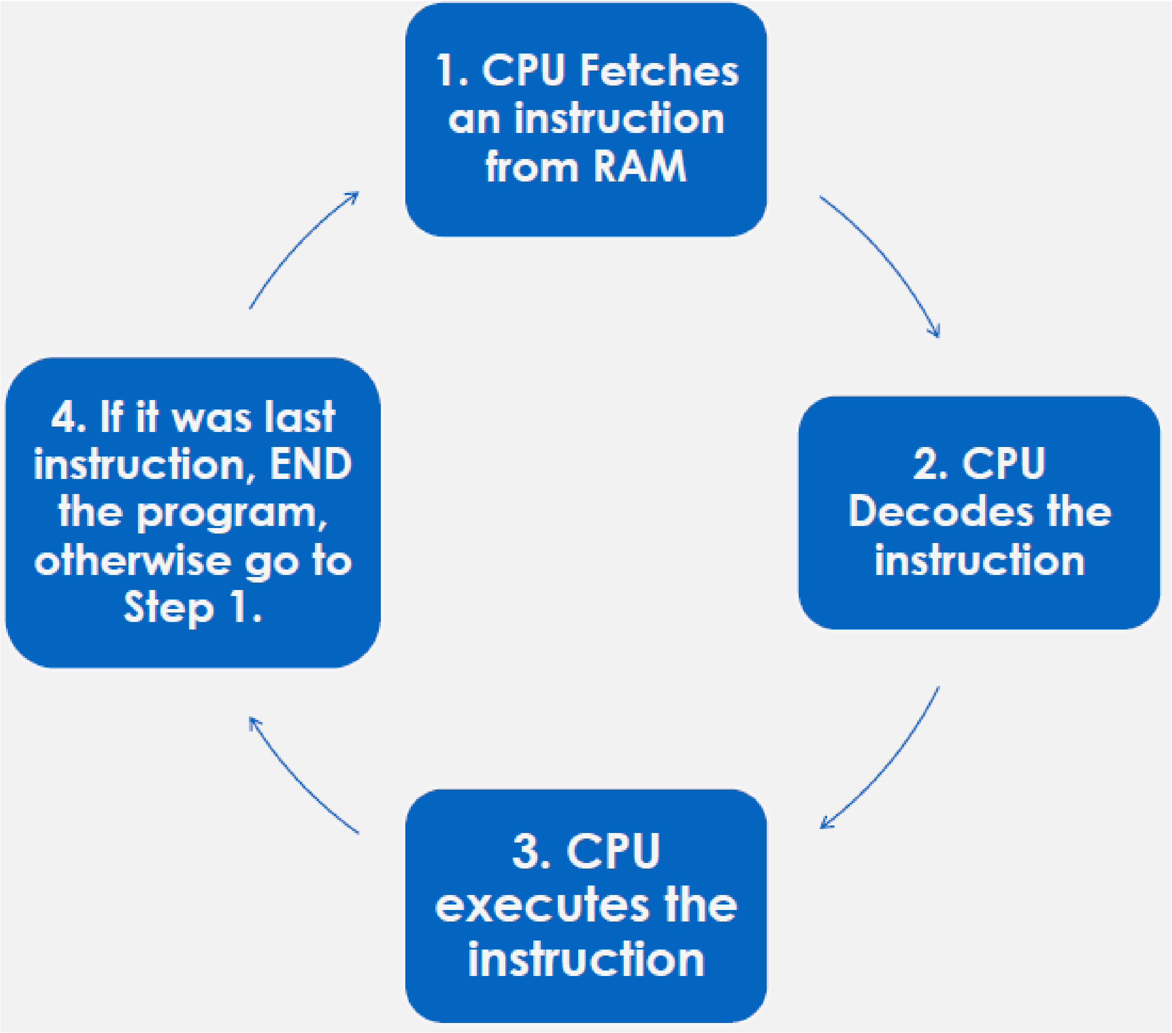
The Anatomy of a Computer

- The CPU performs program control and data processing.
  - ✓ the CPU locates and executes the program instructions;
  - ✓ it carries out arithmetic operations such as addition, subtraction, multiplication, and division;
  - ✓ it fetches data from external memory or devices and places processed data into storage.

Process of running a program



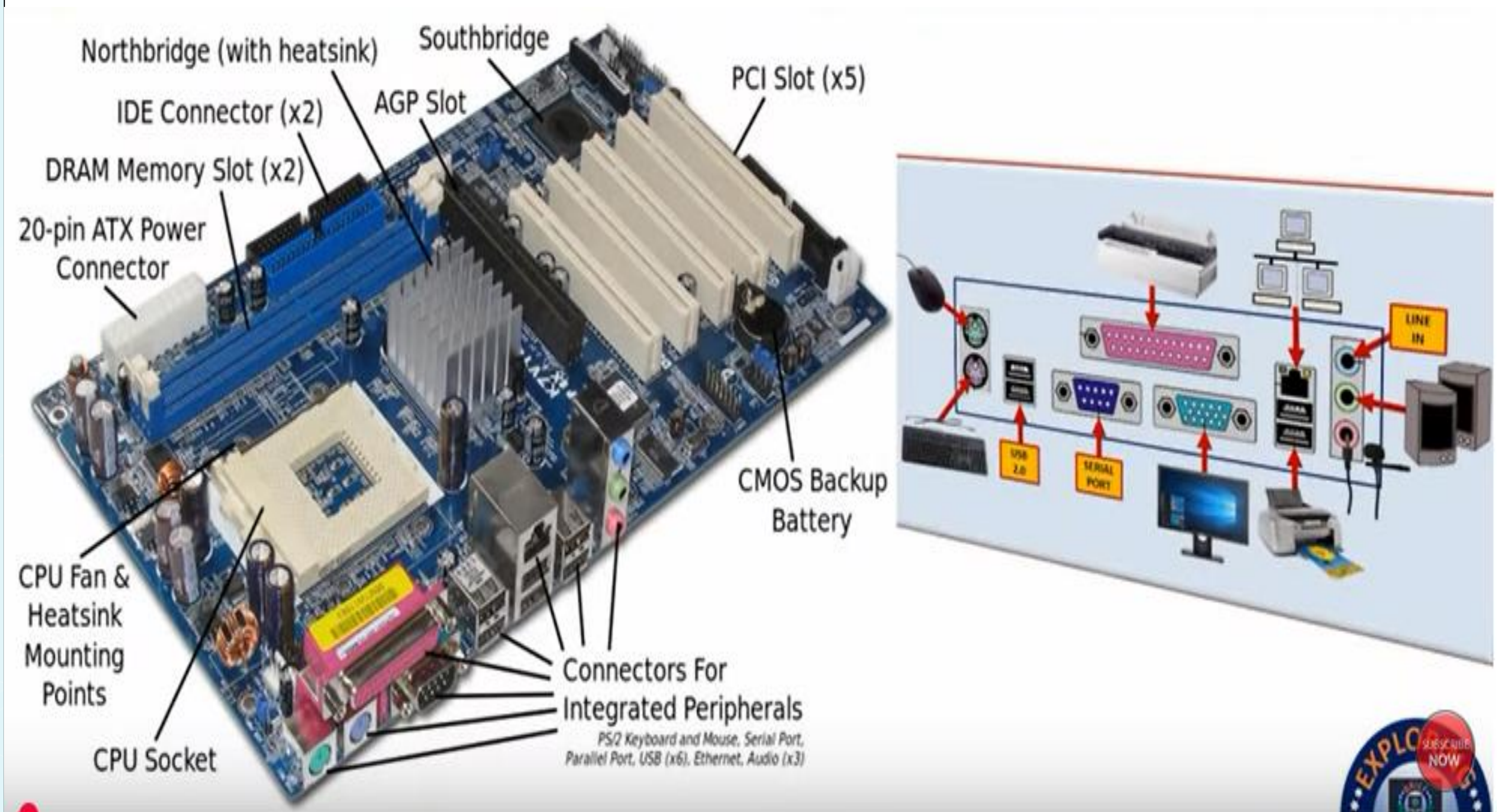
The fetch-execute cycle



❑ Hardware components

- ✓ **Motherboard:** The *main circuit* board that connects all the other components.
- ✓ **Power Supply Unit (PSU):** Provides the necessary *electrical power* to the computer.
- ✓ **Graphics Processing Unit (GPU):** Handles the processing of visual information, especially important for *gaming and graphic design*.
- ✓ **Cooling System:** Prevents the computer from *overheating*, especially the CPU and GPU.

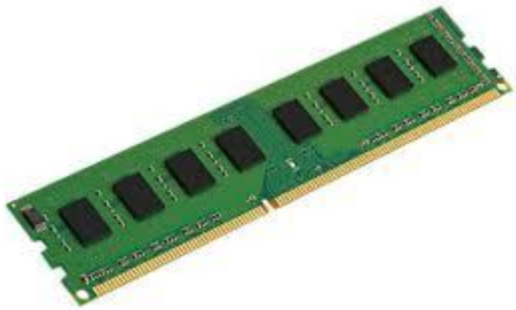
Computer Motherboard



Central Processing Unit

- ✓ The Central Processing Unit (CPU) is usually called either a **CPU** or just a **Processor**.
- ✓ The CPU is the **brain** of the system.
- ✓ It **executes** all the program code from the operating system and the applications the user runs and processing of data.
- ✓ It sends CPU commands to direct the actions of all the other components in the computer.

Random Access Memory (RAM)



- ✓ Random Access Memory (RAM) is a type of computer memory that provides fast, temporary storage for data and instructions that the computer is actively using.
- ✓ It's often referred to as the computer's "short-term memory".
- ✓ RAM is volatile memory, meaning data is lost when the computer is turned off

Hard Disc Drive(HDD)

Solid State Drive (SSD)

Quiz 2

- Where is a program stored when it is not currently running?
- Which part of the computer carries out arithmetic operations, such as addition and multiplication?
- A modern smartphone is a computer, comparable to a desktop computer. Which components of a smartphone correspond to those shown in Figure 1?

The Java Programming Language

- Java was originally designed for programming **consumer devices**, but it was first successfully used to write **Internet applets**.
- Java was designed to be safe and portable, benefiting both Internet users and students.
- Java programs are distributed as instructions for a virtual machine, making them **platform-independent**.
- Java has a very **large library**.
- Focus on learning those parts of the library that you **need** for your programming **projects**.

James Arthur Gosling OC (born 19 May 1955) is a Canadian computer scientist, best known as the **founder** and lead **designer** behind the **Java** programming language.

Cons

- It was not designed to be student friendly
- The language has extensive features –can't all be learnt in one course
- Too many Java updates –**we require** **JDK 24** Long-Term Support (**LTS).**

Quiz

- What are the two most important benefits of the Java language?
- How long does it take to learn the entire Java library?

Programming Environment

A programming environment refers to the comprehensive set of software tools and configurations that a developer utilizes to **write**, **test**, **debug**, and **execute** computer programs. It provides the necessary infrastructure for software development.

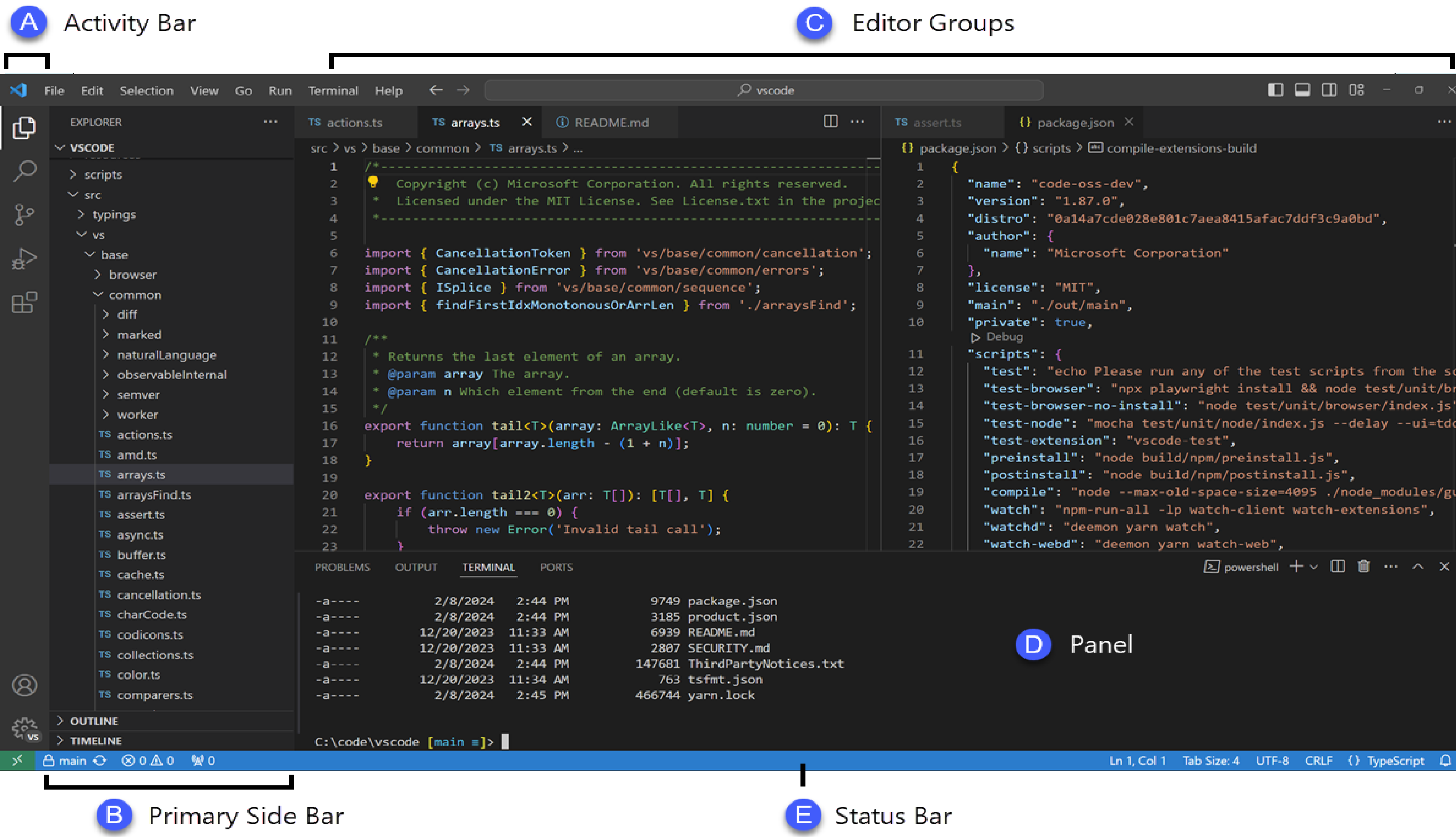
Key components :

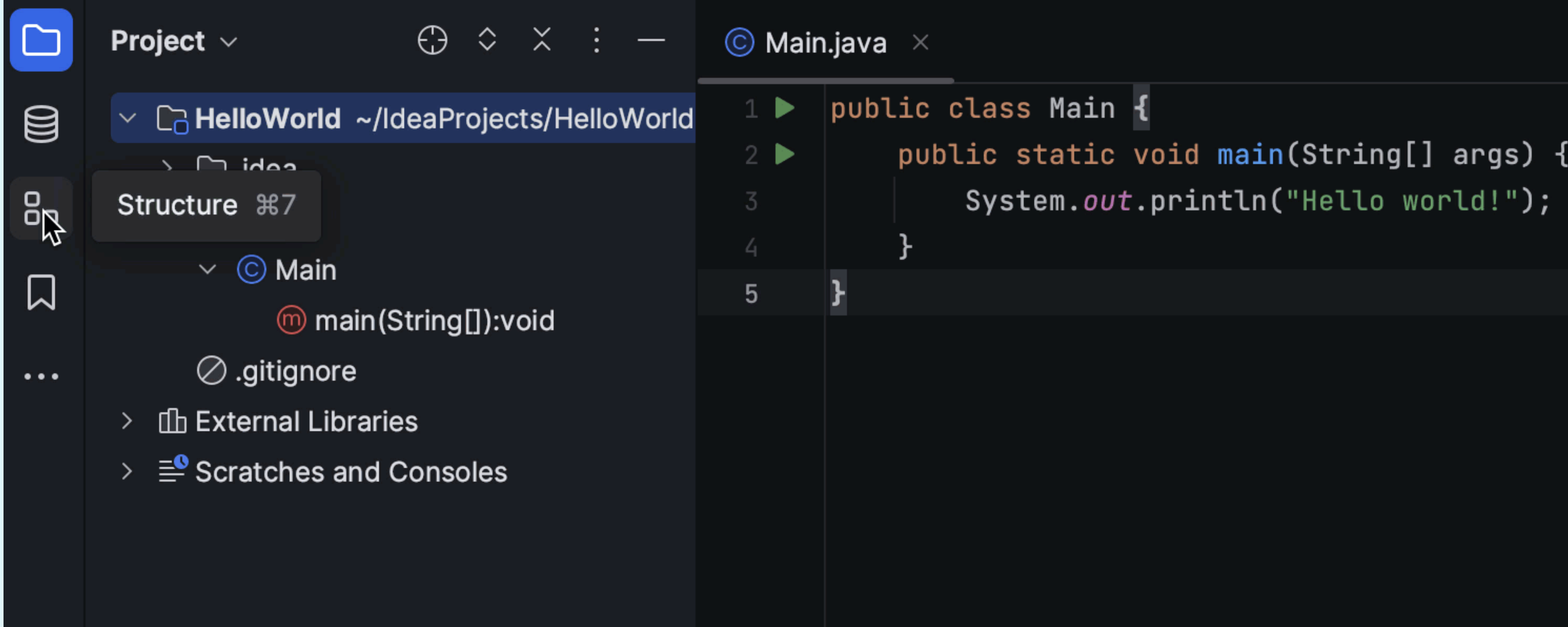
- **Integrated Development Environment (IDE) or Code Editor:**

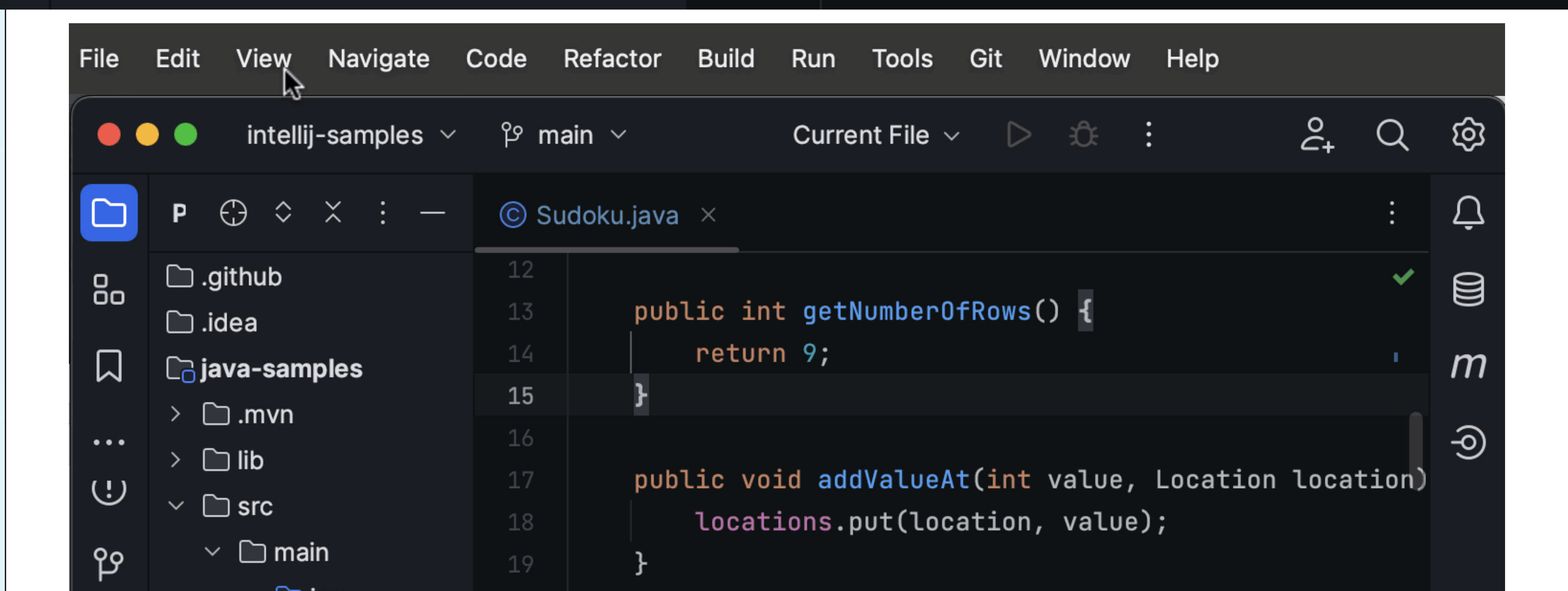
Software that provides a **text editor** for writing code, along with features like **syntax** highlighting, code **completion**, and **project management**.

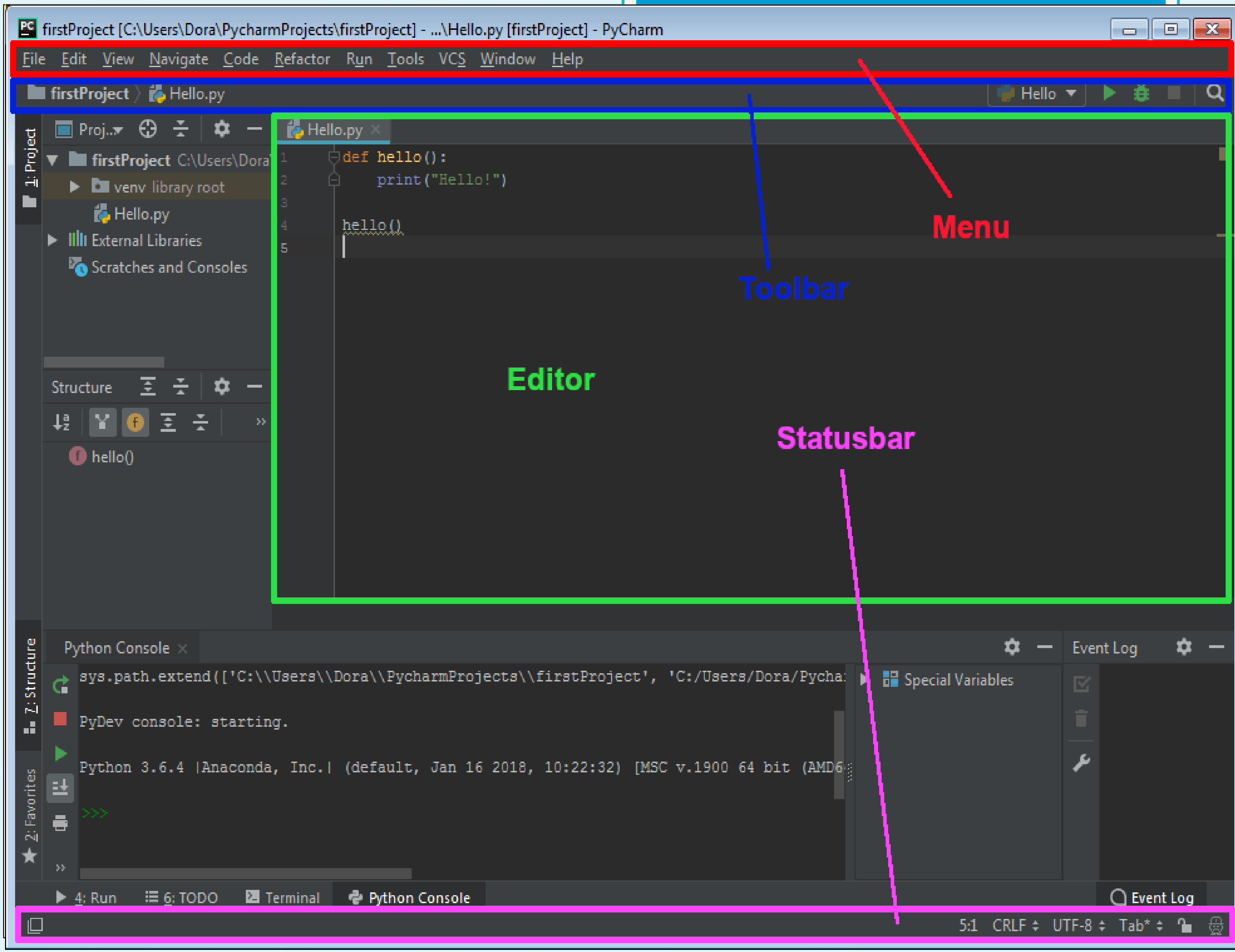
- ✓ IDE: Visual Studio, **IntelliJ IDEA**, Eclipse, PyCharm…
- ✓ Code Editor: Visual Studio Code, Sublime Text, Atom…

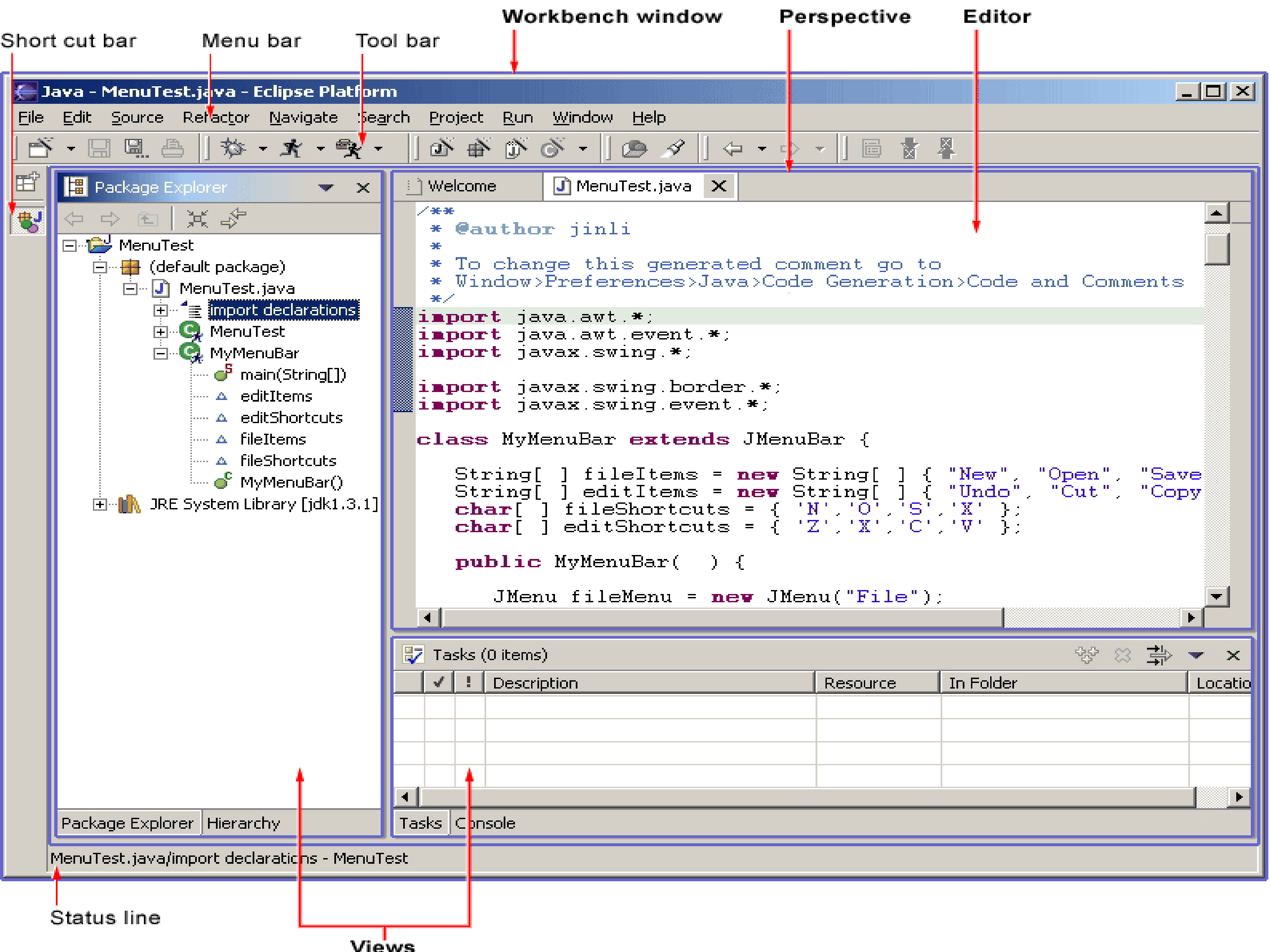
Visual Studio




Project
Main.java
HelloWorld ~/IdeaProjects/HelloWorld
Structure ⌘7
Main
main(String[]):void
.gitignore
External Libraries
Scratches and Consoles
public class Main {
public static void main(String[] args) {
System.out.println("Hello world!");
}
}


File
Edit
View
Navigate
Code
Refactor
Build
Run
Tools
Git
Window
Help
intellij-samples
main
Current File
P
Sudoku.java
.github
.idea
java-samples
.mvn
lib
src
main
public int getNumberOfRows() {
return 9;
}
public void addValueAt(int value, Location location)
locations.put(location, value);
}


firstProject [C:\Users\Dora\PycharmProjects\firstProject] - ...\Hello.py [firstProject] - PyCharm
File Edit View Navigate Code Refactor Run Tools VCS Window Help
firstProject 〉 Hello.py
Hello
Menu
Toolbar
Editor
Statusbar
Proj..
firstProject C:\Users\Dora
venv library root
Hello.py
External Libraries
Scratches and Consoles
Structure
hello()
1: Project
7: Structure
2: Favorites
def hello():
print("Hello!")
hello()
Python Console
sys.path.extend(['C:\\Users\\Dora\\PycharmProjects\\firstProject', 'C:/Users/Dora/Pycha
PyDev console: starting.
Python 3.6.4 |Anaconda, Inc.| (default, Jan 16 2018, 10:22:32) [MSC v.1900 64 bit (AMD6
>>>
Special Variables
Event Log
4: Run
6: TODO
Terminal
Python Console
Event Log
5:1 CRLF UTF-8 Tab*


Workbench window
Perspective
Editor
Short cut bar
Menu bar
Tool bar
Java - MenuTest.java - Eclipse Platform
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer
MenuTest
(default package)
MenuTest.java
import declarations
MenuTest
MyMenuBar
main(String[])
editItems
editShortcuts
fileItems
fileShortcuts
MyMenuBar()
JRE System Library [jdk1.3.1]
Welcome
MenuTest.java
/**
* @author jinli
*
* To change this generated comment go to
* Window>Preferences>Java>Code Generation>Code and Comments
*/
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;
import javax.swing.border.*;
import javax.swing.event.*;
class MyMenuBar extends JMenuBar {
String[ ] fileItems = new String[ ] { "New", "Open", "Save
String[ ] editItems = new String[ ] { "Undo", "Cut", "Copy
char[ ] fileShortcuts = { 'N','O','S','X' };
char[ ] editShortcuts = { 'Z','X','C','V' };
public MyMenuBar( ) {
JMenu fileMenu = new JMenu("File");
Tasks (0 items)
Description
Resource
In Folder
Locatio
Package Explorer
Hierarchy
Tasks
Console
MenuTest.java/import declarations - MenuTest
Status line
Views

Programming Environment

❑ Compiler or Interpreter:

Tools that **translate** human-readable source code into machine-executable code (compiler) or directly **execute** the code (interpreter).

A compiler translates the entire source code **at once** before execution, while an interpreter translates and executes the code **line by line**.

- ✓ Compiler: C, C++, C#, Go
- ✓ Interpreter: Python, Ruby, Perl, JavaScript

Programming Environment

- **Debugger:**

A tool used to **identify and fix errors** (bugs) in the code by allowing step-by-step execution and inspection of program state.

- **Version Control System:**

Software like **Git** or **SVN** used to track changes in code, collaborate with other developers, and manage different versions of a project.

- **Libraries and Frameworks:**

Collections of **pre-written** code that provide functionalities and structures to accelerate development.

Programming Environment

- **Operating System:**

The underlying system software that manages computer hardware and software resources, providing the platform for the programming environment.

- **Command Line Interface (CLI) or Terminal:**

A text-based interface used to interact with the operating system and run various development tools.

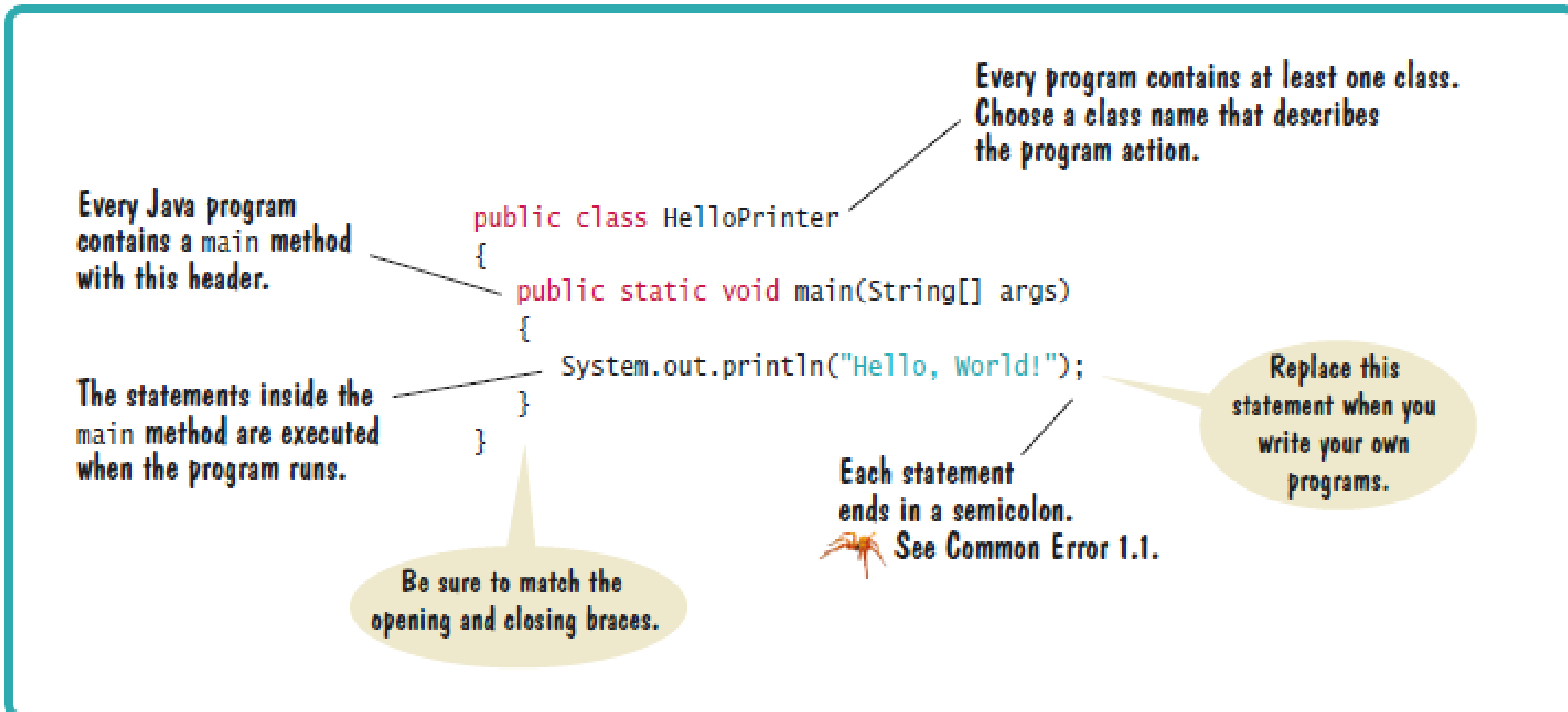
- **Build Automation Tools:**

Tools like Make, Ant, or **Maven** that automate the process of compiling, linking, and packaging software.

Building blocks of a simple program

- **Classes** are the fundamental building blocks of Java programs.
- Every Java application contains a class with a **main method**. When the application starts, the **instructions** in the main method are executed.
- Each class contains **declarations** of methods. Each method contains a **sequence** of instructions.
- A method is called by specifying the method and its **arguments**.
- A **string** is a sequence of characters enclosed in **quotation marks**.

Java Program



Pseudocode

- Pseudocode is a **human-readable**, informal way to describe the steps of an algorithm or a program, using a simplified, **language-agnostic** structure.
- Key Characteristics of Pseudocode:

✓ It doesn't adhere to the strict syntax of any specific programming language. It uses plain English-like statements and conventions.

✓ It's designed to be easily understood by humans,

✓ It prioritizes outlining the **logical flow** of the algorithm or program, rather than the specific syntax of a language.

✓ It serves as a **bridge** between the conceptual idea of a program and its actual implementation in code.

✓ It **cannot** be compiled or executed by a computer

✓ pseudocode has no rigid syntax,

Summary

- ✓ Computers execute very basic instructions in rapid succession.
- ✓ A computer program is a sequence of instructions and decisions.
- ✓ Programming is the act of designing and implementing computer programs.
- ✓ The central processing unit (CPU) performs program control and data processing.
- ✓ Storage devices include memory and secondary storage.
- ✓ Java was originally designed for programming consumer devices, but it was first successfully used to write Internet applets.
- ✓ Java was designed to be safe and portable, benefiting both Internet users and students.
- ✓ Java programs are distributed as instructions for a virtual machine, making them platform-independent.
- ✓ Java has a very large library.

Summary

- ✓ An editor is a program for entering and modifying text, such as a Java program.
- ✓ Java is case sensitive.
- ✓ The Java compiler translates source code into class files that contain instructions for the Java virtual machine.
- ✓ Classes are the fundamental building blocks of Java programs.
- ✓ Every Java application contains a class with a main method. When the application starts, the instructions in the main method are executed.
- ✓ Each class contains declarations of methods.
- ✓ Each method contains a sequence of instructions.
- ✓ A method is called by specifying the method and its arguments.
- ✓ A string is a sequence of characters enclosed in quotation marks.
- ✓ Pseudocode is an informal description of a sequence of steps for solving a problem.

END